Math 514, F13 **Name:** Answer
Quiz 6 **SSN:** xxx-xx-

Instructions: **You must show all of your work to receive a credit for a correct answer.**

Problem 1. (5 points) Suppose that you deposit your money in a bank that pays interest at a nominal rate of 9.6% per year. How long does it take for your money to double if the interest is compounded yearly? Hong long if the interest is compounded continuously?

Compounded yearly:

$$(1+0.096)^n = 2 \Rightarrow n = \frac{\log 2}{\log(1.096)}$$

$$\approx 7.56 \text{ years}$$

Compounded continuously:

$$(e^{0.096})^n = 2 \Rightarrow n = \frac{\log 2}{0.096}$$

$$\approx 7.22 \text{ years}$$

Problem 2. (5 points) How much do you need to invest at the beginning of each of the next 60 months in order to have a value of $200,000 at the end of 60 months, given that the annual nominal interest rate will be fixed at 6% and will compounded monthly?

Monthly interest rate: $r = \frac{0.06}{12} = 0.005$

suppose we invest the amount of A per month. then

$$A + \frac{A}{1+r} + \cdots + \frac{A}{(1+r)^{59}} = \frac{200{,}000}{(1+r)^{60}}$$

$$A\left((1+r)+(1+r)^2 + \cdots + (1+r)^{60}\right) = 200{,}000$$

$$A\,\frac{(1+r)-(1+r)^{61}}{1-(1+r)} = 200{,}000$$

$$A = 200{,}000\left(\frac{-r}{1+r-(1+r)^{61}}\right)$$

$$= 200{,}00 \cdot \frac{0.005}{(1.005)^{61} - 1.005}$$

$$\approx 2852.3.$$